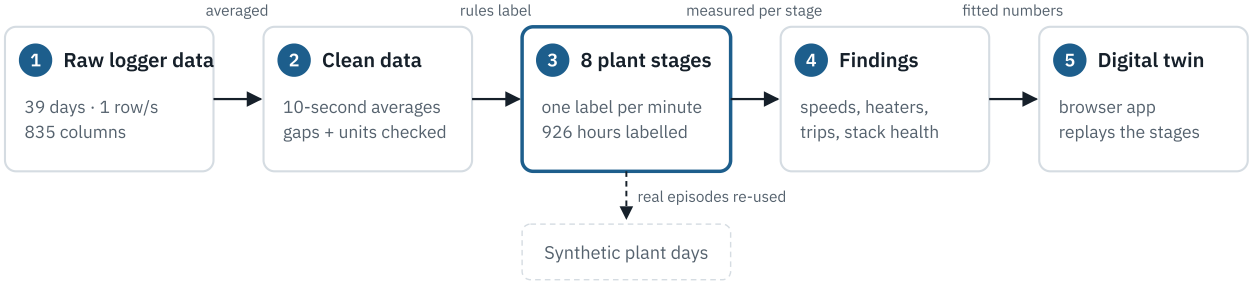
What we did with the plant data

> We took **39 days of logger data** from a 60 kW solid oxide fuel cell plant, gave **every minute one of 8 plant stages**, measured how each part of the plant behaves in each stage, and built a **small digital twin** that replays those stages in a browser.

Why it matters: a stage tells you what the plant is doing right now (warming up, waiting hot, making power, cooling). Once every minute has a stage, you can compare like with like, spot what is unusual, make realistic test data and drive a twin.



The work in five steps. Step 3, the stages, is the key idea: everything after it is measured or replayed stage by stage. The dashed branch is the synthetic-data generator, which re-uses real stretches of the plant's own data.

Read the pages in order

PAGE 2

The starting dataset

What the plant logger gave us, which days, and how it was cleaned.

PAGE 3

Why 8 stages

How every minute got a stage, and why the answer is eight.

PAGE 4

Inside each stage

What is switched on in each stage and how the readings move.

PAGE 5

What the data told us

Heating speeds, the wait for power, heaters, trips, stack health.

PAGE 6

How the twin app works

Knobs → stage rules → one heat equation → 111 readings on screen.

Words used on these pages

WORD	PLAIN MEANING
SOFC	Solid oxide fuel cell: a ceramic cell that turns gas into electricity at about 600 °C. Many cells are piled into a **stack**.
Forming gas (N_2 + H_2)	Nitrogen with a little hydrogen. Blown over the fuel side of the hot stack so it does not rust (oxidise) while it waits.
Natural gas (NG)	The real fuel. A reformer first turns it into a hydrogen-rich gas.
Air heater (EAPH)	Electric heater that warms the air going into the stack. Like a hair-dryer for the stack.
Recuperators (APH)	Heat exchangers: hot exhaust warms the fresh air. No power, they work by themselves whenever air flows.
Gas-line heaters	Small electric heaters (ERH, EAGH, EAOH 1–3) that keep the fuel-side pipes hot.
Trip	A safety cut. The plant stops itself when a reading crosses a limit.

If you remember one thing: Every minute of the plant now has one of 8 stages, and the findings and the twin are all built on those stages.

The starting dataset

> The POC dataset is the plant's own logger: **one text file per day, one row every second, 835 columns** (temperatures, flows, pressures, heater currents, voltages and control commands). We received 39 such days.

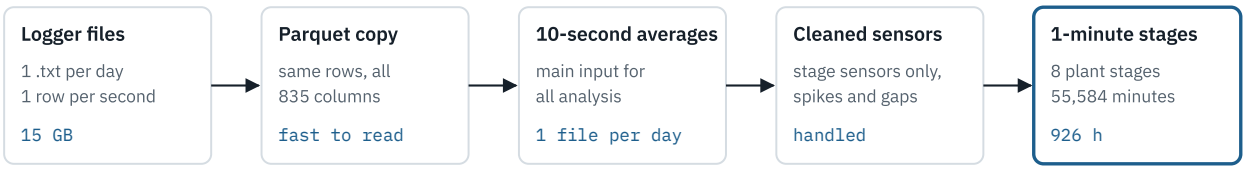


When the plant was logged. The days come in five groups between 30 Oct 2025 and 30 Mar 2026, so stages that span a gap (a long cool-down) are followed across the missing days.

The numbers

WHAT	VALUE	NOTE
Logged days	39	30 Oct 2025 → 30 Mar 2026
Rows per day	86,400	one per second
Columns	835	sensors + controller set-points and commands
Raw size	15 GB	whole data folder with all copies: 28 GB
Sensors on the plant's sensor list	156	each one traced to a logged column and rated
Hours given a stage	926 h	13 h had no readable data

What was done first



From raw files to labelled minutes. Each step writes its own copy, so the next step never touches the raw logger files.

- **Made the data quick to read.** The text files were copied to a compact table format (parquet) and averaged to 10-second steps.
- **Checked every sensor on the list.** Each of the 156 listed sensors was matched to a logged column, and marked as useful, flat or missing.
- **Worked out what the confusing tags mean.** For example `FIRC 1.2` is the reformer air, the natural gas is `FIRC 1.4`, and the signal called "FIRC 1.1.3" is really a temperature-driven valve.
- **Separated commands from measurements.** Operator set-points change first; the plant follows 1–18 minutes later. That told us where stage borders really start.
- **Read the plant documents.** The stage definitions and the safety map (in `01_docs/`) gave the first stage list and every trip limit.

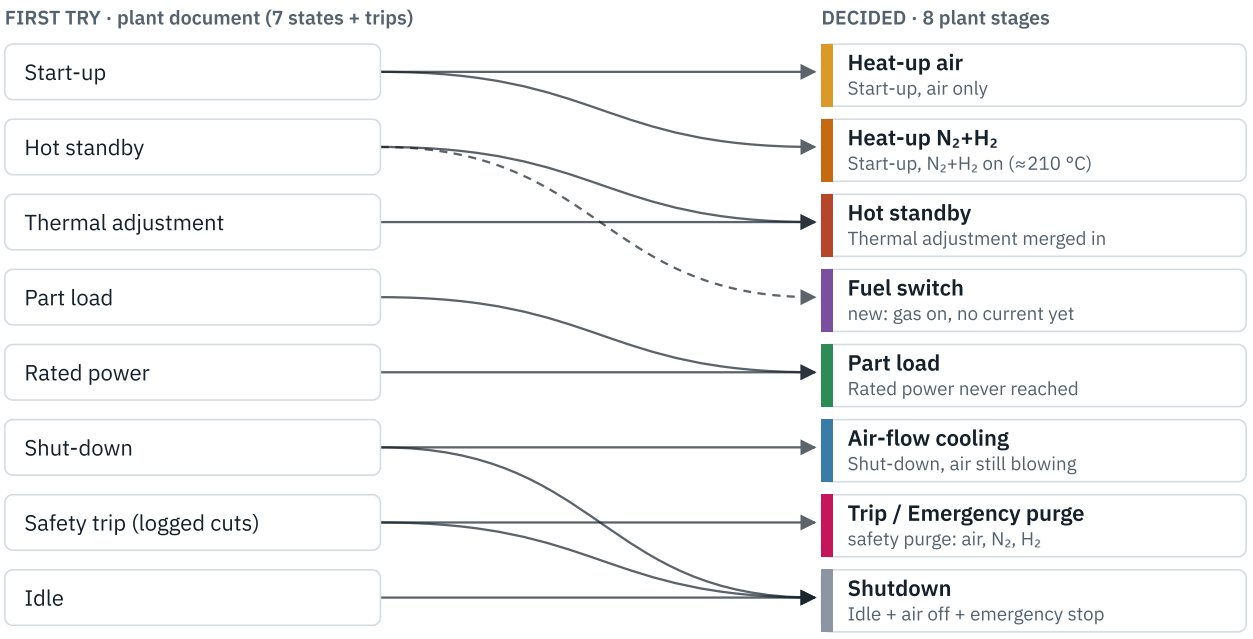
Where it lives
`raw days: 02_data/raw/ · 10 s averages: 02_data/downsampled/`
`cleaning: 03_data_analysis/preprocessing.ipynb · stage file:`
`03_data_analysis/plant_stages/plant_stages_1min.parquet`

If you remember one thing: 39 days of one-second logger data were cleaned into 10-second averages, which feed everything else.

Why there are 8 stages

> A stage is defined as **a set of supplies and devices that are switched on**. When we looked at which supplies (air, forming gas, natural gas, current) and heaters were on, minute by minute, the plant only ever showed **eight different sets**. So there are eight stages.

The first try named stages after the stack temperature, as the plant document does. That gave names that did not say what was running, a "Rated power" stage that never happened, and no place for the minutes where gas was on but no power came out yet.



How the first stage list became the 8 decided stages. Arrows show where each old state went; the dashed arrow is the new Fuel switch stage, cut out of hot standby where natural gas is already on. The second line in each box says why.

The decisions behind the eight

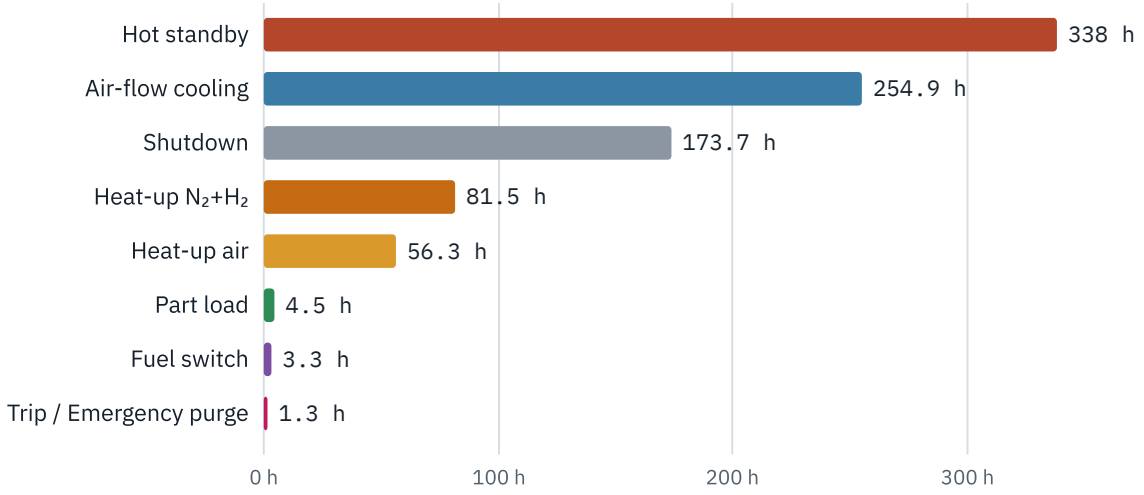
DECISION	REASON SEEN IN THE DATA
Split start-up into **Heat-up air** and **Heat-up N_2+H_2**	Different supplies. Forming gas comes on when the stack passes about 210 °C; this rule holds on 96.1 % of minutes.
Merge **Thermal adjustment** into **Hot standby**	Same devices on; it is only a re-heat or cool-back of a hot, waiting stack.
Add **Fuel switch**	Natural gas is on but the current is below 1 A. It is the 5–30 min bridge (median 10) between waiting and making power.
Merge **Rated power** into **Part load**	Full power was never reached: the highest string current was about 16 A of roughly 30 A rated.
Split shut-down by **air flow**	With air blowing the stack cools 10–20 times faster than with air off, so they behave as two stages.
Merge Idle, air-off cooling and emergency stop into **Shutdown**	In all three only the controls are on.
Keep **Trip / Emergency purge** separate	A safety-only set: air, N_2 and H_2 on, everything else off, for 15–17 min.

How a minute gets its stage



The rule is checked top to bottom; the first “yes” wins. Trips are checked first because safety overrides everything. The inputs are measured flows, current and the stack temperature from the 10-second data.

Where the plant spent its time



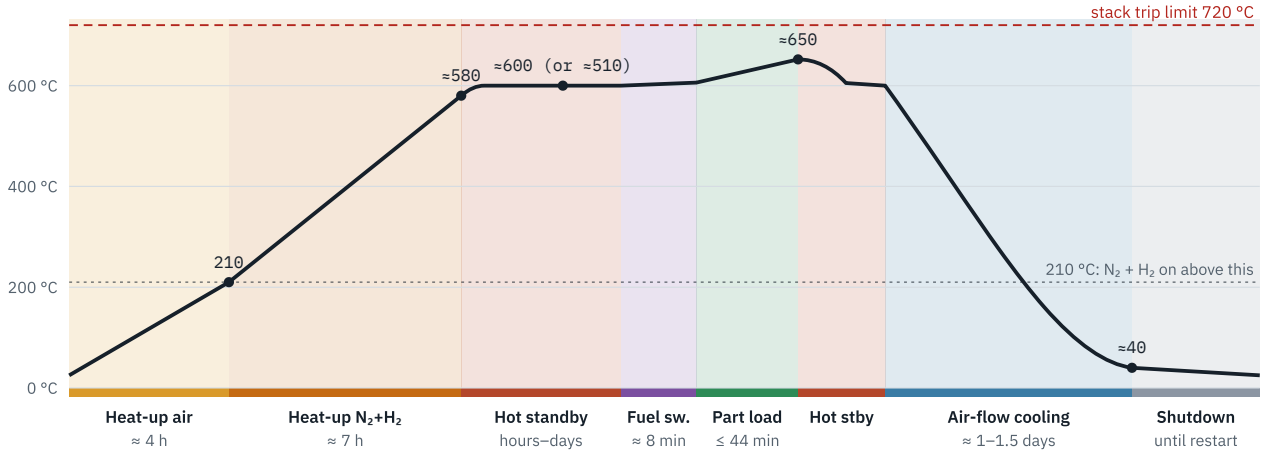
Hours in each stage over the 926 labelled hours. The plant spent most of its time waiting hot or cooling down; it made power for only 4.5 hours in total.

Where it lives
`rules: 03_data_analysis/scripts/plant_stages.py (built on operating_states.py) · notebook: operating_states.ipynb §12`

If you remember one thing: Eight stages, because the plant shows exactly eight different combinations of what is switched on.

Inside each stage

A normal run follows one chain: **heat-up air → heat-up N_2+H_2 → hot standby → fuel switch → part load → back to hot standby → air-flow cooling → shutdown**. The trip purge is the safety exit. Each step switches a supply on or off, and the readings follow.



The stack temperature through one typical run, using the plant's median values. Widths are not to scale; the typical length of each stage is printed under it. The stack never came near the 720 °C trip limit except in part load.

What is switched on

Pick a stage to highlight its column and jump to its card. Values are plant medians (flows in ln/min, i.e. normal litres per minute).

- Heat-up air
- Heat-up N_2+H_2
- Hot standby
- Fuel switch
- Part load
- Air-flow cooling
- Shutdown
- Trip / Emergency purge

Part	Heat-up air	Heat-up N_2+H_2	Hot standby	Fuel switch	Pa
Cathode air (2 streams)	● 6000	● 6000	● 6000	● 6000–8000	9
Electric air heater EAPH	● 13 kW	● 20 kW	● 10 kW	● 11 kW	9
Recuperators APH (passive)	●	●	●	●	
Gas-line heaters ERH, EAGH, EAOH 1–2	● 3 A	● 3 A	● 3 A	● 5 A	3-
Recycle heater EAOH 3	● 8 A	● 8 A	● 8 A	● 12 A	
Forming gas N_2 1200 + H_2 60	○	●	●	○	
Natural gas + reformer air	○	○	○	● 80 + 343	137
Reformer + gas recycle hot	○	○	○	●	
After-burner burning	○	○	○	◐ starting	
Stack current to loads	○	○	○	○ < 1 A	6-
Controls	●	●	●	●	

● on ◐ partly or only sometimes ○ off Heaters below full power count as on.

The eight stages in detail

1 Heat-up air

Start-up with hot air only

The stack is cold. Two air streams blow 6000 ln/min through it and the electric air heater warms that air. Nothing flows on the fuel side yet.

Stack temperature	25 → 210 °C (median 159)
Air into stack	≈ 204 °C
Air heater power	≈ 13 kW
Gas-line heaters	≈ 3 A each, recycle heater ≈ 8 A
Stack voltage	≈ 0 V (no gas on the fuel side)

How values change: Stack warms steadily, about +42 °C/h (limit 120 °C/h). The heaters follow the air: they come on 0–9 min after the air starts.

`Typical visit ≈ 4 h (longest 9.5 h). Next: Heat-up N₂+H₂ in 11 of 14 cases.`

2 Heat-up N_2+H_2

Start-up with forming gas

Once the stack passes about 210 °C, nitrogen with a little hydrogen is sent over the fuel side to protect it. The air heater now works hardest.

Stack temperature	≈ 230 → 560 °C (median 463)
Forming gas	N_2 1200 + H_2 60 ln/min
Air heater power	≈ 20 kW (its highest)
Air into stack	≈ 500 °C
Stack voltage	jumps to ≈ 420–470 V per string

How values change: Stack keeps warming at about +39 °C/h. The voltage appears as soon as hydrogen reaches the cells, even with no current drawn. The after-burner is not burning yet.

`Typical visit ≈ 7 h. Next: Hot standby in 9 of 15 cases.`

3 Hot standby

Hot and ready, no power

The stack is held hot with forming gas on, waiting for the order to make power. This is where the plant spent most of its time.

Stack temperature	held at ≈ 510 or ≈ 600 °C (two groups)
Change	about ±1 °C/h, flat
Air	6000 ln/min, air in ≈ 500 °C
Air heater power	≈ 10 kW
Stack voltage	≈ 430 V per string, ≈ 110 V per stack

How values change: Readings stay flat. Re-heats and cool-backs (the old “thermal adjustment”) are started by operator commands. The compressor fills its tank near the end.

```
338 h in total; visits from minutes to 80 h. Next: Fuel switch (17
of 25) or Air-flow cooling (6).
```

4 Fuel switch

Natural gas on, no current yet

Forming gas goes off and natural gas plus reformer air come on. The reformer and the after-burner light up. The stack is ready but no current is drawn yet.

Natural gas	≈ 80 ln/min + reformer air 343
Reformer catalyst inlet	≈ 756 °C
After-burner	≈ 643 °C
Stack temperature	≈ 600 °C, still rising ≈ +22 °C/h
Stack voltage	431 → 467 V in about 3.5 min

How values change: The stack never settles here. Gas heat pushes it up while the voltage climbs to its open-circuit value. Gas-line heaters step up to ≈ 5 A, recycle heater ≈ 12 A.

```
Typical visit ≈ 8 min (max 29). Gas-on to first current: median 10
min. Next: Part load (10 of 19) or back to Hot standby (7).
```

5 Part load

Making power

Current flows from the five stack strings into the loads. More natural gas and more air are needed. Full rated power was never reached.

String current	median 6–7 A (max ≈ 16 A, rated ≈ 30 A)
Natural gas	≈ 137 ln/min + reformer air 312
Air	≈ 9000 ln/min, air in ≈ 600 °C
Stack temperature	≈ 604 → 652 °C
After-burner	≈ 741 °C; string voltage ≈ 455 V

How values change: The stack heats fast because the current itself makes heat. No run lasted long enough to settle; at 720 °C the plant trips.

```
10 runs, 4.5 h in total, longest 44 min. Next: Air-flow cooling (3),
Hot standby (2), Fuel switch while unloading (2), Trip (2).
```

6 Air-flow cooling

Shutting down, air still blowing

Gas and heaters are off. Air keeps blowing to carry the heat out at a safe speed. Forming gas stays on while the stack is above 210 °C.

Air	6000 ln/min (sometimes 8000–10000)
Air heater, gas-line heaters	off
Forming gas	on while stack > 210 °C
Stack temperature	falls from ≈ 600 to ≈ 40 °C (median 242)
Stack voltage	fades as the hydrogen stops

How values change: Cooling speed depends on the air: about −19 °C/h at 6000 ln/min and −45 °C/h at 8000–10000 ln/min.

```
255 h in total. Next: Shutdown (12 of 23) or a restart into Heat-up
air (8).
```

7 Shutdown

Everything off except the controls

Air, gases, heaters and current are all off. Covers a cold plant, a warm plant left to cool with no air, and the first minutes after an emergency stop.

Air and gases	0
Heaters	off
Stack temperature	median 44 °C (ambient up to ≈ 190 °C)
Change	about −2 °C/h when still warm
Stack voltage	≈ 0 V

How values change: Very slow drift towards room temperature. The hot box holds its heat for days without air.

`174 h in total. Next: Air-flow cooling (14), Hot standby (7) or Heat-up air (4).`

8 Trip / Emergency purge

Safety stop

A reading crossed a safety limit and the plant cut itself off. Only air, nitrogen and hydrogen keep flowing to flush the stack safely.

Air	6000 ln/min
Forming gas	N_2 1200 + H_2 60 ln/min
Everything else	off
Stack temperature	≈ 630–690 °C
Length	15–17 min

How values change: Most trips came from the exhaust safety reading TIRZ 8.1.10 going just over its limit seconds before the cut (6 of 9 trips).

`6 purges, 1.3 h in total. Next: Shutdown (3), Air-flow cooling (2) or Hot standby (1).`

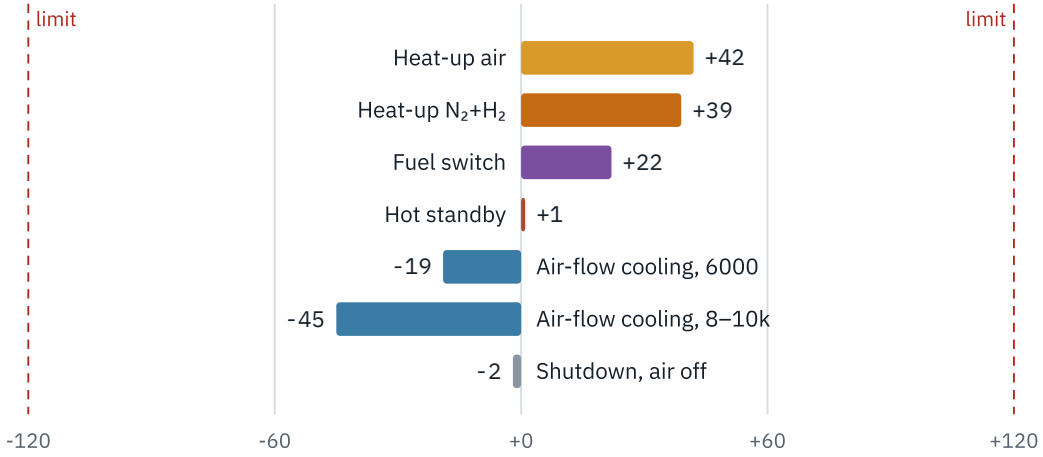
Sources

`stage file: 03_data_analysis/plant_stages/ · per-stage sensor medians: 05_twin_2d/data/sensors.js · speeds: 03_data_analysis/slopes_jerk/`

If you remember one thing: Each stage is one switch-on pattern: air first, then forming gas, then natural gas, then current, and back off in reverse.

What the data told us

With every minute labelled, the plant's behaviour could be measured stage by stage. The headline: **the plant mostly waited hot or cooled down, made power for only 4.5 hours, warmed up well inside its limits, and tripped mostly on one exhaust temperature reading.**



How fast the stack temperature moves in each stage, in °C per hour (median of the plant runs). Every stage stays far inside the 120 °C/h limit for heating and cooling. Part load is left out: its runs were too short to give a steady speed.

Nine findings

FINDING 1

Full power never reached

The plant made power for 4.5 h in 10 runs, the longest 44 min, at about 6–7 A of ≈ 30 A rated. Stretching the data to 60 kW suggests ≈ 51 kW (46–56) would be reached.

FINDING 2

Heating and cooling speeds

Heat-up runs at +42 / +39 °C/h, a third of the 120 °C/h limit. A cold start to ≈ 490 °C takes 11.6–12 h. Cooling is −19 °C/h at 6000 ln/min air, −45 at 8000–10000, −2 with air off.

FINDING 3

The wait for power

From natural gas on to the first current: 5–30 min, median 10. During that wait the stack never settles (+22 °C/h) and the voltage rises 431 → 467 V in 3.5 min.

FINDING 4

Heaters follow the air

The gas-line heaters switch with the air flow, not with the stack temperature, and lead the stack by 25–30 min in heat-up. Their current changed with the date: ≈ 3 A until 12 Mar, ≈ 5 A on 13–24 Mar.

FINDING 5

Commands come first

Set-points, valve and fuel commands change 1–18 min before each stage border. Every re-heat or cool-back of the hot stack (11 of 11) started with an operator command.

FINDING 6

Why the plant tripped

9 trips. In 6 of them the exhaust safety reading TIRZ 8.1.10 went just over its limit seconds before the cut. The stack itself peaked at 704 °C, under its 720 °C limit.

FINDING 7

Stack health

In hot standby each stack shows ≈ 110 V, steady with the hydrogen flow. Stack 4.3 weakened from 91 to 79 % of its neighbours during March.

FINDING 8

Availability

In March the plant was available 40 % of the time (28 % by the clock), with a trip on average every 14 hours of running.

FINDING 9

Efficiency

No efficiency signal is logged. Calculated from gas in and power out it rises from 10 to 33 % as the power goes up.

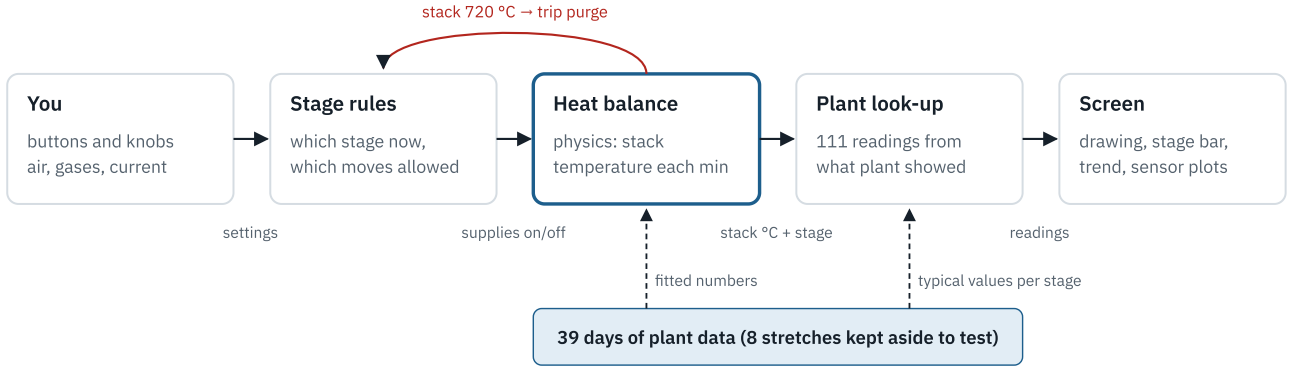
Questions each notebook answered

TOPIC	NOTEBOOK (03_DATA_ANALYSIS/)
Start-up recipe, rates, gases, timing	`startup_analysis.ipynb`
Re-heats and cool-backs of a hot stack	`thermal_adjustment_analysis.ipynb`
Current ramps, cell voltage, exhaust trips	`part_load_analysis.ipynb`
Cool-down speeds, forced cooling	`shutdown_analysis.ipynb`
How each of the 156 sensors behaves per stage	`sensors_by_stage.ipynb`
Stack temperature vs gas-line heaters	`heater_temperatures.ipynb`
Heating/cooling slopes and bends at stage borders	`slopes_jerk.ipynb`

If you remember one thing: The plant spent its life warming, waiting and cooling; making power was short and rare, so part load is the least known stage.

How the twin app works

The twin is a small app that runs on your own computer. You press stage buttons or turn knobs; **one heat equation** works out the stack temperature minute by minute, and the **other 111 readings** are looked up from what the real plant showed in that stage at that temperature.



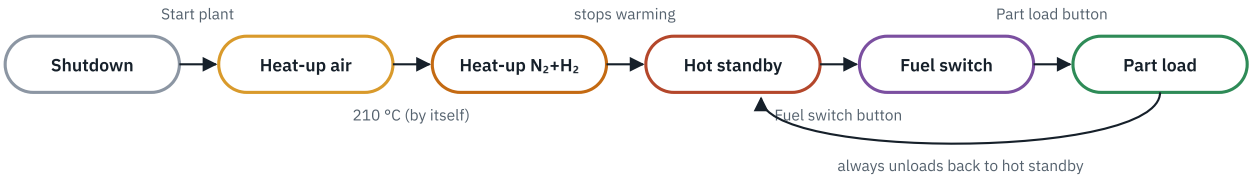
The five parts of the twin. Only the stack temperature is real physics (blue outline). Everything else on screen is the plant's own typical behaviour, taken from the 39 days.

The heat equation, in words

```
stack warms per hour = hot air in + heaters + gas heat + current heat − heat lost to the room
```

PART	FITTED ON THE PLANT DATA
Hot air in	Swaps heat with the stack at 0.561 per hour when warming it, 0.316 when cooling it; stronger with more air
Recuperator	Gives the fresh air 88 % of the stack's heat at 6000 ln/min (83 % at 8000+)
Air heater	Pushes the air-in towards its target at the plant's 60 °C/h, up to 40 kW
Gas heat	0.370 °C/h for each ln/min of natural gas
Current heat	6.61 °C/h for each ampere per string
Heat lost	0.0121 per hour × (stack − 25 °C)

How the stages move in the app



Shut down → Air-flow cooling at 9000 ln/min → Shutdown when stack < 40 °C

Trip (hot standby / part load, or 720 °C) → 16-min purge → Hot standby

Emergency stop (any time) → everything off at once → Shutdown

The twin only allows the moves the plant makes. Heat-up steps happen by themselves; power needs two buttons in order; leaving part load always goes back to hot standby. The app opens in Shutdown and waits for Start plant.

How close it is to the real plant

TEST	RESULT
Replay of 8 plant stretches the fit never saw	`stack error 56 → 17.5 °C`
10 plant part-load runs	`mean stack miss 4.2 °C`
Cold start to ≈ 490 °C	`twin 11.4 h · plant 11.6–12.0 h`
Cooling out of hot standby (5 runs)	`miss 12.2 → 5.1 °C`
Readings blending at stage changes	`error 19.9 → 13.3 % of the move`

Limits: only the stack temperature is computed; other readings are typical values, not predictions per part. The plant never ran part load longer than 44 min, so where it would settle is unknown. Cooling at 9000 ln/min is still too slow in the twin.

Open it
`double-click 05_twin_2d/app.bat (own Edge/Chrome window, no server) · desktop icon: install_shortcut.bat once`
`press` **Start plant** `to run a cold start · click any part or line to see its sensors and their 24 h plots`

If you remember one thing: The twin is physics for the stack temperature and the plant's own memory for everything else.